
Artelys Knitro in PowSyBl Open Load Flow: a Resilient Optimization-Based AC Load Flow

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ABSTRACT

We present a resilient optimization-based formulation of the AC load flow problem, implemented with Artelys Knitro inside the open-source PowSyBl (Power System Blocks) framework. Unlike the classical Newton-Raphson method, which may fail on ill-conditioned or infeasible cases without explaining why, our approach introduces bounded-sign slack variables on the active power, reactive power and target-voltage equations, and penalizes them in the objective. The retained penalty is a **pure ℓ_1 norm**: it is exact, it keeps the objective linear in the slacks, and it removes the structural incentive to spread a violation over many buses, which is what makes the returned solution diagnosable. We then derive a physically grounded methodology to set the three remaining weights ($\omega_P, \omega_Q, \omega_V$) from business quantities: a reactive anchor, an MW/MVAr equivalence driven by the DC active imbalance, and a kV/MVAr equivalence driven by the short-circuit power at the bus. The model was validated through stress tests on IEEE networks with controlled perturbations, and on real data. Where Newton-Raphson fails to converge, the proposed model consistently returns feasible, physically interpretable solutions whose active slacks localize the inconsistency.

Keywords. Operations Research · AC Load Flow · Nonlinear Programming · Exact Penalty

1 Introduction

Solving the load flow and the Optimal Power Flow (OPF) problems is vital to guarantee the security and the efficiency of modern power systems. The OPF ensures that all operational limits — voltage, line flow, generator capacity — are respected, thus maintaining system stability and preventing violations that could lead to breakdowns or equipment damage. It also provides valuable insight for sensitivity analyses, helping operators understand how changes in system conditions, such as demand fluctuations or the integration of renewable energy sources, affect network performance.

Newton-Raphson is the reference numerical algorithm to solve the load flow equations. It is fast and robust on well-posed cases, but it suffers from two practical limitations on real data:

- (i) when the data are inconsistent (conflicting voltage setpoints, unrealistic active imbalance after a merging process, erroneous shunt sections), the underlying square system is simply **infeasible** and no iterative scheme can converge;

- (ii) when it fails, it returns a divergence flag and nothing else: the user has no indication of **where** the problem is.

This work investigates a different route: reformulating the load flow as a nonlinear optimization problem, solved with Artelys Knitro, in which infeasibility is absorbed by explicit slack variables. The solution is then always available, and the set of non-zero slacks acts as a diagnosis of the dataset.

The document is organized as follows. Section 2 recalls the physical and mathematical formalism and the reference square formulation. Section 3 presents the resilient model, discusses the choice of the penalty function and justifies the retained pure ℓ_1 form. Section 4 derives the weight-setting methodology, which is a direct consequence of that choice. Section 5 and Section 6 describe the implementation and the validation protocol.

2 Formalism

2.1 Mathematical representation of power systems

The physical electrical network is approximated by a mathematical model, represented as an undirected graph $\mathcal{G}$, where nodes model important locations of the actual grid (generation points, load points), and where edges model the behavior of electrical lines, such as resistance and reactance effects.

Nodes are named **buses**. At every bus $k \in \mathcal{N}$, we denote by $v_k = |V_k|e^{j\theta_k}$ the complex voltage, with V_k and θ_k the unknown voltage magnitude and angle. These two quantities are determined by solving the power flow equations defined in Section 2.3.

A generator (production) or a load (consumption) can be linked to a bus, thus setting a production or consumption target at this bus. All buses with a voltage-regulating generator are referred to as **PV buses**, the others as **PQ buses**. For simplicity, we consider that a bus is either PV or PQ. At a PV bus, the active power P and the voltage magnitude V are specified; at a PQ bus, the active power P and the reactive power Q are specified. We denote by $\mathcal{N}_{\text{PV}}$ and $\mathcal{N}_{\text{PQ}}$ the corresponding sets, with $\mathcal{N}_{\text{PV}} \cup \mathcal{N}_{\text{PQ}} =: \mathcal{N}$.

Edges link two buses and represent an electrical branch with its parameters. We denote by $\mathcal{E}$ the set of edges, and by $\text{Ngh}(i)$ the set of buses adjacent to i .

2.2 Π model of a branch

Let $i, j \in \mathcal{N}$. A commonly adopted model to represent a transmission line or a branch linking two buses is the Π model: a series impedance and two shunt admittances, one at each end of the series impedance.

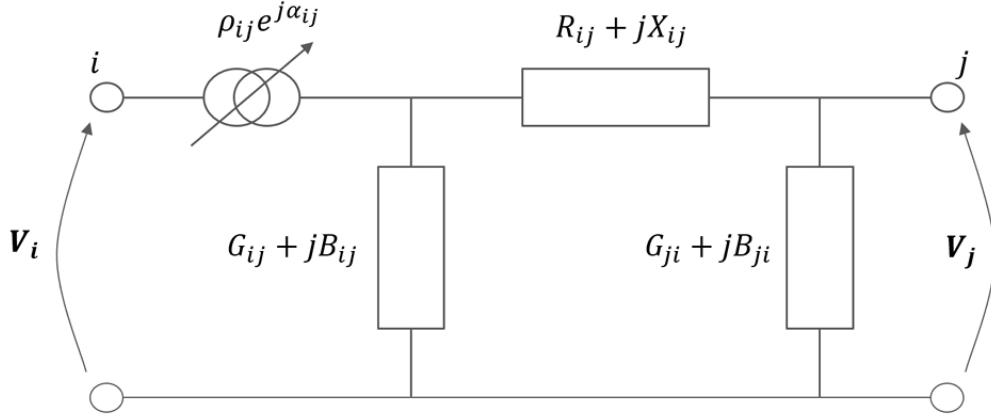


Figure 1: Π model of a transmission line

We denote by:

- $Z_{ij} = R_{ij} + jX_{ij}$ the complex impedance, representing the resistance and the reactance of the line, connected in series. The resistance accounts for the real power loss in the conductor; the reactance is associated with the magnetic field surrounding the conductor.
- G_{ij} (resp. G_{ji}) the conductance and B_{ij} (resp. B_{ji}) the susceptance on side i (resp. side j). They represent the real power loss caused by current leakage — currents flowing to the ground — and by corona discharge.
- ρ_{ij} the magnitude ratio and α_{ij} the phase shift between side j and side i , which together form a transformer. Transformers control magnitudes and phases, and through them, power. By default there is no transformer on a branch, hence $\rho_{ij} = 1$ and $\alpha_{ij} = 0$; in that case the branch is a line, linking buses at the same voltage level. Otherwise it is a transformer.

The active and reactive powers flowing from bus i to bus j are given by the following nonlinear, non-convex expressions:

$$p_{ij}(V, \theta) := \rho_{ij} V_i V_j Y_{ij} \sin(\theta_i - \theta_j + \alpha_{ij} - \Xi_{ij}) + \rho_{ij}^2 V_i^2 (G_{ij} + Y_{ji} \sin(\Xi_{ij})) \quad (1)$$

$$q_{ij}(V, \theta) := -\rho_{ij} V_i V_j Y_{ij} \cos(\theta_i - \theta_j + \alpha_{ij} - \Xi_{ij}) + \rho_{ij}^2 V_i^2 (-B_{ij} + Y_{ji} \cos(\Xi_{ij})) \quad (2)$$

where $R + jX = \frac{1}{Y_{ij}} e^{j(\frac{\pi}{2} - \Xi_{ij})}$.

2.3 Power flow equations

We arbitrarily designate a generator bus as the **slack bus**, denoted $s \in \mathcal{N}$. The slack bus maintains the overall active power balance: P is not specified there, which makes it able to absorb any imbalance between production and consumption inherent to the network (including the power dissipated by the Joule effect).

Removing the equation in P would make the system under-determined, since the slack bus would only carry its magnitude equation. Instead of setting P at this bus, we fully set the voltage by also fixing the phase. Since the power equations 1 and 2 only depend on phase **differences**, the phase at the slack bus can be set to zero; every other bus voltage is then expressed relative to the slack bus.

The system is then fully determined:

- at every bus $i \in \mathcal{N}$ there are two unknowns, V_i and θ_i , hence $2 \times |\mathcal{N}|$ variables;

- we therefore need $2 \times |\mathcal{N}|$ non-redundant equations.

At the slack bus s :

$$\theta_s = 0, \quad V_s = V_s^{\text{ref}} \quad (3)$$

At PV buses, $\forall i \in \mathcal{N}_{\text{PV}} \setminus \{s\}$:

$$P_i = \sum_{j \in \text{Ngh}(i)} p_{ij}(V, \theta), \quad V_i = V_i^{\text{ref}} \quad (4)$$

At PQ buses, $\forall i \in \mathcal{N}_{\text{PQ}}$:

$$P_i = \sum_{j \in \text{Ngh}(i)} p_{ij}(V, \theta), \quad Q_i = \sum_{j \in \text{Ngh}(i)} q_{ij}(V, \theta) \quad (5)$$

Equations in V are called **target V** equations, setting the operating target of PV buses. Equations in P and Q are **flow conservation** constraints, maintaining injection equilibrium at every bus. Together they form a $2 \times |\mathcal{N}|$ nonlinear, non-convex system, commonly called the **power flow equations**.

2.4 Reference (square) formulation

For the sake of clarity, the reference model solved by Newton-Raphson can be written as a feasibility problem with a null objective:

$$\begin{aligned} \min_{V, \theta} \quad & 0 \\ \text{s.t.} \quad & P_i - \sum_{j \in \text{Ngh}(i)} p_{ij}(V, \theta) = 0, \quad \forall i \in \mathcal{N} \setminus \{s\} \\ & Q_i - \sum_{j \in \text{Ngh}(i)} q_{ij}(V, \theta) = 0, \quad \forall i \in \mathcal{N}_{\text{PQ}} \\ & V_i - V_i^{\text{ref}} = 0, \quad \forall i \in \mathcal{N}_{\text{PV}} \\ & \theta_s = 0 \\ & V_i, \theta_i \in \mathbb{R}, \quad \forall i \in \mathcal{N} \end{aligned} \quad (6)$$

This formulation has **no** degree of freedom: either the system admits a solution, or the solver fails.

2.5 Nomenclature

Symbol	Meaning
$\mathcal{N}, \mathcal{E}$	sets of buses and branches
$\mathcal{N}_{PV}, \mathcal{N}_{PQ}$	PV (voltage-regulating) and PQ buses
s	slack bus
V_i, θ_i	voltage magnitude (p.u.) and angle (rad) at bus i
p_{ij}, q_{ij}	active / reactive flow from i to j , see 1 and 2
$s_{P_i}^\pm, s_{Q_i}^\pm, s_{V_i}^\pm$	non-negative slacks on the P , Q and target- V equations
$\omega_P, \omega_Q, \omega_V$	penalty weights of the objective
S_{base}	power base, 100 MVA
$S_{\text{cc}}, I_{\text{cc}}$	short-circuit power / current at a bus
γ	voltage stiffness, $\gamma = S_{\text{cc}}/S_{\text{base}}$
ρ_{PQ}	MW/MVAr equivalence ratio
η	voltage de-prioritization factor

Table 1: Main notation.

3 The resilient model

3.1 Principle: relaxing with signed slacks

The idea is to add slack variables on the P , Q and target- V equations so that the feasible set is never empty, and to penalize these variables in the objective so that they stay at zero whenever the physical problem admits a solution.

Each equation $c(V, \theta) = 0$ becomes $c(V, \theta) - s^+ + s^- = 0$ with $s^+, s^- \geq 0$, so that the residual is $\varepsilon = s^+ - s^-$ and its absolute value is $s^+ + s^-$ **provided** that $s^+ s^- = 0$. This complementarity does not need to be imposed: with any penalty that is increasing in $s^+ + s^-$, subtracting $\min(s^+, s^-)$ from both variables keeps the point feasible and strictly decreases the objective, so any optimal solution satisfies $\min(s^+, s^-) = 0$.

3.2 Choice of the penalty function

Let ε denote the residual of a relaxed equation. Three candidates were considered.

Quadratic (ℓ_2), $h(\varepsilon) = \varepsilon^2$. Smooth and easy for the solver, but it **diffuses**: since the cost is strictly convex, spreading a total violation S over n buses costs $n \cdot (S/n)^2 = S^2/n$, which strictly decreases with n . The solver therefore prefers to smear an inconsistency over the whole network rather than expose it where it is. This destroys the diagnostic value of the slacks, which is the main purpose of the model, and it does so **independently of the weights** ω .

Huber. The historical model used an adaptive Huber-type loss, quadratic near zero and linear beyond a threshold τ , augmented with a kink at the origin to avoid a vanishing marginal penalty for small residuals:

$$h(\varepsilon) := \begin{cases} \frac{1}{2}\varepsilon^2 + \lambda|\varepsilon| & \text{if } |\varepsilon| \leq \tau, \\ \tau(|\varepsilon| - \frac{1}{2}\tau) + \lambda|\varepsilon| & \text{otherwise.} \end{cases} \quad (1)$$

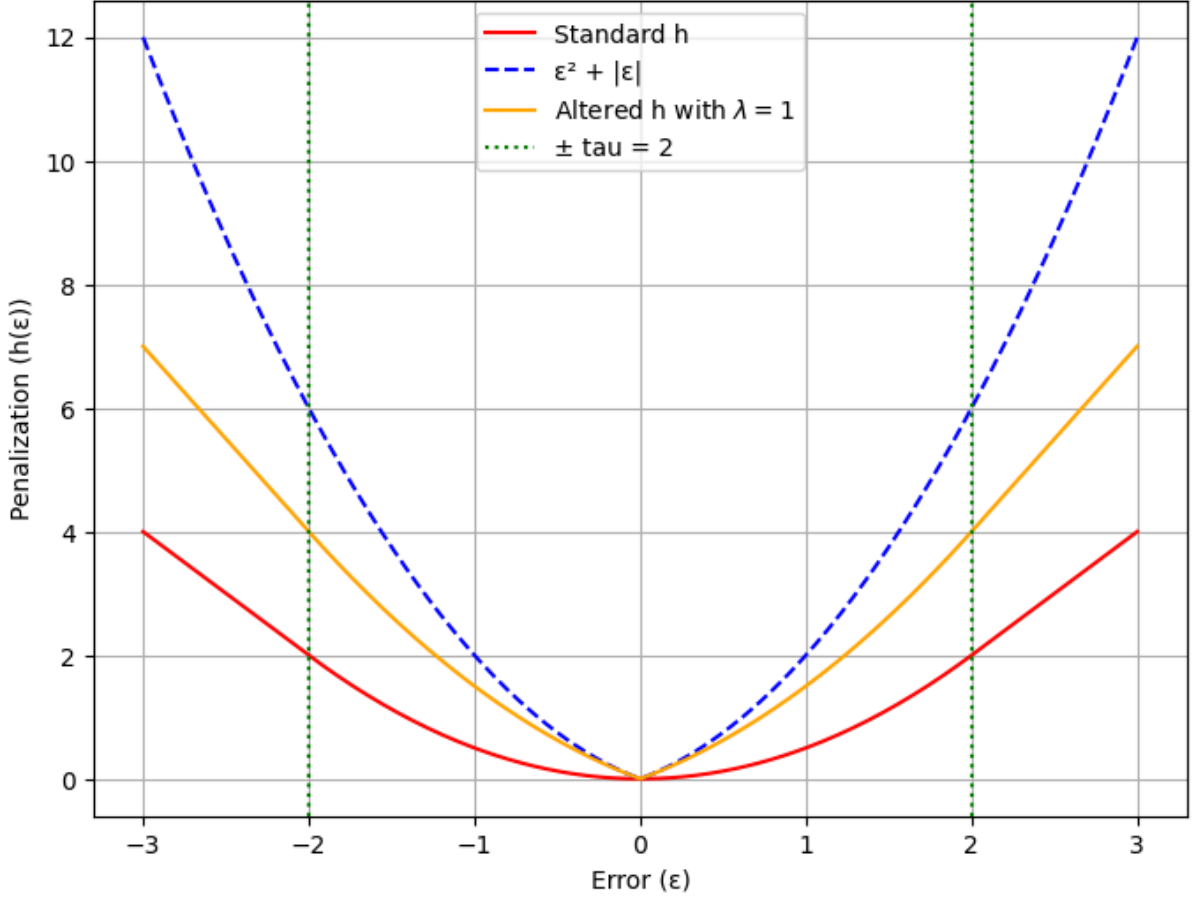


Figure 2: Huber's loss function (quadratic near the origin, linear beyond τ).

It requires the calibration of one threshold τ per constraint type, in addition to the weights, and it retains the diffusive behaviour in the quadratic regime.

Mixed $\ell_2 + \ell_1$. The model that was actually implemented used $h(\varepsilon) = \mu\varepsilon^2 + \lambda|\varepsilon|$ with $\mu = 1$, $\lambda = 3$ and unit weights. The marginal cost of the ℓ_2 term, $2\mu\varepsilon$, equals that of the ℓ_1 term, λ , at

$$\varepsilon^* = \frac{\lambda}{2\mu} = 1.5 \text{ p.u.} = 150 \text{ MW} \quad (S_{\text{base}} = 100 \text{ MVA}). \quad (2)$$

In other words, on the whole range of realistic violations, **the implemented penalty was already behaving like an ℓ_1 norm**: the quadratic term only took over beyond 150 MW, exactly in the regime where diffusion is most harmful.

Retained choice: pure ℓ_1 . We therefore drop the quadratic term entirely and keep

$$h(s^+, s^-) = s^+ + s^-. \quad (3)$$

This formalizes the behaviour the model already had in the useful range, and removes the diffusion mechanism in the large-violation range. It has four further consequences that structure the rest of this document.

- (i) **Exactness.** The ℓ_1 penalty is an exact penalty function: for weights above the magnitude of the optimal multipliers, the relaxed problem and the original one share the same solutions. Here the property is even stronger, see Section 3.4.
- (ii) **Linearity.** With the s^+/s^- splitting, the objective is **linear**. The nonlinearity and non-convexity of the problem are entirely carried by the flow equations; the Hessian of the Lagrangian comes only from the constraints, which is numerically favorable for an interior-point or SQP method.
- (iii) **No incentive to diffuse.** With an ℓ_1 norm, distributing a given total violation over n buses costs the same as concentrating it on one. The penalty is neutral, and the arbitration is left to the physics: spreading an imbalance over electrically distant buses requires additional transits, hence additional losses and additional residuals elsewhere. In practice, the solver concentrates. This is the classical argument behind ℓ_1 relaxations of cardinality objectives — the ℓ_1 norm is the convex envelope of ℓ_0 on bounded sets — and it is what makes the solution **readable**: a short list of non-zero slacks pointing at the faulty data.
- (iv) **Only ratios matter.** Scaling all weights by the same positive constant leaves the argmin unchanged. The calibration problem is therefore two-dimensional, not three-dimensional: one weight can be fixed by convention and the other two are **relative prices**.

Numerical note. The objective being linear in the slacks, the problem is degenerate by construction: several slack distributions may share the same optimal cost. If this turns out to hinder convergence or reproducibility, a vanishing quadratic regularization $\varepsilon \sum (s^+ - s^-)^2$ with ε several orders of magnitude below the smallest weight can be added to select a unique solution without reintroducing the diffusion effect.

3.3 The optimization model

Using 3, the resilient AC load flow is the following nonlinear program:

$$\begin{aligned}
& \min_{V, \theta, s^\pm} \quad \omega_P \sum_{i \in \mathcal{N} \setminus \{s\}} (s_{P_i}^+ + s_{P_i}^-) + \omega_Q \sum_{i \in \mathcal{N}_{PQ}} (s_{Q_i}^+ + s_{Q_i}^-) + \omega_V \sum_{i \in \mathcal{N}_{PV}} (s_{V_i}^+ + s_{V_i}^-) \\
& \text{s.t.} \quad \sum_{j \in \text{Ngh}(i)} p_{ij}(V, \theta) - P_i^{\text{ref}} - s_{P_i}^+ + s_{P_i}^- = 0, & \forall i \in \mathcal{N} \setminus \{s\} \\
& \quad \sum_{j \in \text{Ngh}(i)} q_{ij}(V, \theta) - Q_i^{\text{ref}} - s_{Q_i}^+ + s_{Q_i}^- = 0, & \forall i \in \mathcal{N}_{PQ} \\
& \quad V_i - V_i^{\text{ref}} - s_{V_i}^+ + s_{V_i}^- = 0, & \forall i \in \mathcal{N}_{PV} \\
& \quad \theta_s = 0 \\
& \quad 0.5 \leq V_i \leq 1.5, & \forall i \in \mathcal{N} \\
& \quad s_{P_i}^+, s_{P_i}^- \geq 0, & \forall i \in \mathcal{N} \setminus \{s\} \\
& \quad s_{Q_i}^+, s_{Q_i}^- \geq 0, & \forall i \in \mathcal{N}_{PQ} \\
& \quad s_{V_i}^+, s_{V_i}^- \geq 0, & \forall i \in \mathcal{N}_{PV} \\
& \quad \theta_i \in \mathbb{R}, & \forall i \in \mathcal{N}
\end{aligned} \tag{4}$$

Compared with the historical formulation 6, the feasible set is never empty: any (V, θ) satisfying the bounds can be completed by slacks. The solver therefore always returns a point, and the question becomes the **quality** and the **interpretability** of that point.

3.4 The weights are inert on healthy networks

The objective of 4 is non-negative and vanishes if and only if all slacks are zero, that is, if and only if (V, θ) solves the original power flow equations. Consequently:

If the load flow problem is feasible, then for any strictly positive weights the global optima of 4 are exactly the solutions of the original system 6.

The weights play **no role** on well-posed cases: they cannot bias the result, only the numerical path towards it. They matter only when the data are inconsistent, where they arbitrate **which** equation is relaxed and by how much. Two practical consequences:

- a non-regression benchmark on healthy networks validates the implementation and the numerics, but says nothing about the calibration;
- the calibration must be evaluated on the perturbed instances of Section 6, by looking at **where** the slacks land, not at the objective value.

3.5 Discarded alternative: normalization by tolerances

An alternative discussed during the study was to normalize each slack by a characteristic tolerance $\delta_P, \delta_Q, \delta_V$ representing an operationally acceptable deviation, making the objective dimensionless, e.g. 5% for active power, 10% for reactive power, 1% for voltage. Under a pure ℓ_1 penalty, this is not an alternative but a **reparametrization**: dividing the slack by δ_t is exactly equivalent to setting $\omega_t = 1/\delta_t$. The methodology of Section 4 can therefore be read either way, as a choice of relative prices or as a choice of tolerances; we keep the weights formulation because the physical derivations below produce prices directly.

4 Setting the objective weights

Because the retained penalty is a pure ℓ_1 norm, the objective is a **linear price system**: ω_t is the cost of one per-unit of violation of a constraint of type t , and only the ratios between the three prices matter. The calibration is therefore organized as a cascade starting from a single anchor.

4.1 Step 1: anchor — reactive weight

We fix the reactive weight as the reference. It represents the cost of one MVar of deviation, expressed in objective units:

$$\omega_Q := 1 \quad (\text{by convention}) \quad (1)$$

Every other quantity is expressed relative to this anchor. Any other value would give the same optimal solutions, by the scale invariance noted above.

4.2 Step 2: active weight — MW/MVar equivalence

The weight ω_P expresses how many MVar of deviation one MW of deviation is “worth”. We write it as an equivalence ratio ρ_{PQ} :

$$\omega_P = \rho_{PQ} \cdot \omega_Q = \rho_{PQ} \quad (2)$$

The ratio is modulated by the **global active imbalance** of the case, computed before solving under DC assumptions:

$$\Delta_P = \sum P_{\text{gen}} - \sum P_{\text{load}} - \hat{L} \quad (3)$$

where $\hat{L}$ estimates the losses. A first approximation is $\hat{L} \approx \alpha \sum P_{\text{load}}$ with $\alpha \approx 2\%-3\%$. More precisely, under DC assumptions the active transit on line (i, j) is $P_{ij}^{\text{DC}} = (\theta_i - \theta_j)/x_{ij}$ in p.u., and the Joule losses on that line, with $|I| = P_{ij}^{\text{DC}}/V$, are

$$L_{ij} = r_{ij} \frac{(P_{ij}^{\text{DC}})^2}{V^2} \approx r_{ij} (P_{ij}^{\text{DC}})^2 \quad (4)$$

the last approximation holding for $V \approx 1$ p.u. Total losses are $\hat{L} = \sum_{(i,j) \in \mathcal{E}} L_{ij}$.

We choose ρ_{PQ} inversely proportional to $|\Delta_P|$: the better the active balance of the dataset, the more suspicious a residual active slack is, hence the more it should be penalized. Conversely, a case that is grossly unbalanced — typically after merging national networks with inconsistent border exchanges — **has** to place active slacks somewhere, and forcing them to be expensive would only push the infeasibility onto the voltage targets. With $P = \sum P_{\text{gen}}$ the total generation:

$$\rho_{PQ} = \min \left(\max \left(\frac{P}{10|\Delta_P|}, \rho_{\min} \right), \rho_{\max} \right), \quad \rho_{\min} = 10, \quad \rho_{\max} = 1000 \quad (5)$$

The ratio exceeds 10 as soon as $|\Delta_P| < P/100$, i.e. as soon as the imbalance is below 1% of the generation. The floor $\rho_{\min}$ guarantees that active deviations always remain at least an order of magnitude more expensive than reactive ones; the cap $\rho_{\max}$ avoids a degenerate price when $\Delta_P \rightarrow 0$ and keeps the objective well scaled.

The form and the bounds of 5 are heuristic; near balance, its sensitivity is dominated by the error on $\hat{L}$, and the calibration is addressed in the experimental protocol of Section 7.

4.3 Step 3: voltage weight — kV/MVAr equivalence

The physical link between voltage and reactive power is given by the sensitivity of a bus connected to the rest of the network through an equivalent impedance. Consider a bus at voltage V connected to a voltage source E (the rest of the network) through an impedance $Z = R + jX$. The apparent power injected at the bus is

$$S = V \cdot I^* = V \left(\frac{E - V}{Z} \right)^* \quad (6)$$

In polar coordinates ($V = V\angle\delta$, $E = E\angle 0$), the reactive power reads

$$Q = \frac{VE \cos \delta - V^2}{X} - \frac{VE \sin \delta}{R} \quad (7)$$

On a transmission network, $R \ll X$ (the R/X ratio is typically of the order of 0.1) and angle deviations are small ($\cos \delta \approx 1$, $\sin \delta \approx 0$), so

$$Q \approx \frac{V(V - E)}{X} \quad (8)$$

Differentiating around the operating point ($V \approx E \approx V_{\text{nom}}$):

$$\Delta Q \approx \frac{V_{\text{nom}} \Delta V}{X} \implies \frac{\Delta V}{V_{\text{nom}}} \approx \frac{\Delta Q \cdot X}{V_{\text{nom}}^2} \quad (9)$$

The sensitivity of voltage to reactive power is inversely proportional to the **stiffness** of the network. In per-unit, this defines

$$\gamma := \frac{V_{\text{nom}}^2}{X \cdot S_{\text{base}}} = \frac{1}{X_{\text{p.u.}}}, \quad \text{so that} \quad s_V^{\text{p.u.}} = \frac{s_Q^{\text{p.u.}}}{\gamma} \quad (10)$$

Two ways of estimating γ were considered.

Approach 1 — typical reactance. The linear reactance of a 400 kV overhead line is of the order of $x_l \approx 0.3 \, \Omega/\text{km}$. For a length $L \approx 150 \, \text{km}$ (order of magnitude of an inter-substation 400 kV link in mainland France), $X_{\text{line}} = 0.3 \times 150 = 45 \, \Omega$. With $Z_{\text{base}} = V_{\text{nom}}^2/S_{\text{base}} = 400^2/100 = 1600 \, \Omega$, we get $X_{\text{p.u.}} \approx 0.03$ and $\gamma \approx 33$. This estimate is crude: it accounts for a single line, ignoring both the number of lines connected to the bus and the network beyond the first neighborhood.

Approach 2 — short-circuit power (recommended). The three-phase short-circuit current is $I_{\text{cc}} = V_{\text{nom}}/(\sqrt{3}X)$ (phase-to-neutral voltage over the Thévenin impedance, under $R \ll X$), hence $S_{\text{cc}} = \sqrt{3}V_{\text{nom}}I_{\text{cc}} = V_{\text{nom}}^2/X$. Both approaches thus give the same expression:

$$\gamma = \frac{V_{\text{nom}}^2}{X \cdot S_{\text{base}}} = \frac{S_{\text{cc}}}{S_{\text{base}}} \quad (11)$$

The numerical difference comes only from the X used: approach 1 approximates it by the reactance of a single typical line, whereas I_{cc} embeds the true Thévenin impedance of the whole network seen from the bus. Approach 2 is therefore exact by construction, and I_{cc} can be computed quickly by Powsybl before the load flow, bus by bus if a local γ is wanted.

Voltage level	I_{cc} (kA)	S_{cc} (MVA)	γ	$\omega_V = \eta\gamma$
420 kV	63	45 830	458.3	45.8
245 kV	50	21 218	212.2	21.2
145 kV	40	10 046	100.5	10.0
63 kV	31.5	3 437	34.4	3.4

Table 2: Orders of magnitude of γ per voltage level ($S_{\text{base}} = 100 \, \text{MVA}$, $\eta = 0.1$).

Finally, since we would rather move a voltage setpoint — often wrong, and always adjustable — than inject reactive power on units that are already at their limits, we apply a de-prioritization factor $\eta = 0.1$:

$$\omega_V = \eta \cdot \gamma \cdot \omega_Q = \eta \cdot \gamma \approx 0.1 \cdot \frac{S_{\text{cc}}}{S_{\text{base}}} \quad (12)$$

Interpretation. γ measures the voltage stiffness of the network. A strongly meshed network (large S_{cc} , small X) has a large γ : there, one p.u. of voltage deviation corresponds to a lot of reactive power, so relaxing a voltage target is expensive. A radial network has a small γ . The factor η tilts the arbitration towards relaxing V targets rather than reactive balance.

4.4 Summary

Weight	Formula	Depends on	Physical meaning
ω_Q	1	anchor (convention)	cost per MVar of deviation
ω_P	ρ_{PQ}	Δ_P, P (DC pre-computation)	MW/MVar equivalence
ω_V	$\eta\gamma = \eta S_{cc}/S_{base}$	network (I_{cc}, V_{nom}), η	kV/MVar equivalence

Table 3: The three weights of the ℓ_1 model and their determinants.

The inputs of the system are:

- **Engineering choices:** η (voltage de-prioritization), and the bounds ρ_{min}, ρ_{max} of 5.
- **Network quantities:** $V_{nom}, S_{base}, S_{cc}$ (or I_{cc}) — read or computed automatically.
- **Imbalance signal:** Δ_P, P — computed before solving under DC assumptions, determining ρ_{PQ} .

Note that the ℓ_2 -related parameters of the previous formulation ($\tau_P, \tau_Q, \tau_V, \lambda, \mu$, the ℓ_1/ℓ_2 crossover thresholds $\Delta P_{seuil}, \Delta Q_{seuil}, \Delta V_{seuil}$, and the outer weights w_K) all disappear with the pure ℓ_1 choice. Six degrees of freedom are reduced to two effective ones.

4.5 Indicative numerical values

With $S_{base} = 100$ MVA, $V_{nom} = 145$ kV, $I_{cc} = 40$ kA (hence $S_{cc} = 10\,046$ MVA and $\gamma = 100.5$), $\eta = 0.1$, a total generation $P = 1000$ MW and a residual imbalance $|\Delta_P| = 1$ MW, giving $\rho_{PQ} = 1000/10 = 100$:

Weight	Value	Reading
ω_Q	1	1 MVar of reactive deviation = 1 unit
ω_P	100	1 MW of active deviation = 100 MVar
ω_V	10.0	1 p.u. of voltage deviation = 10 p.u. of reactive

Table 4: Indicative weights on a 145 kV case.

The ratio $\rho_{PQ} = 100$ expresses a strong priority on preserving the active balance when the P balance is nearly closed. The factor $\eta = 0.1$ encourages the solver to adjust voltage setpoints rather than to correct in reactive power. Reading the last line concretely: relaxing a voltage target by 0.01 p.u. costs 0.1 objective units, i.e. the equivalent of 10 MVar — a deliberate design choice, consistent with the fact that voltage setpoints in real datasets are the least reliable input.

5 Implementation and testing framework

We implemented the resilient load flow model in Java, as part of the `powsybl-open-loadflow-knitro-solver` project¹. This implementation is integrated into the PowSyBl² framework, an open-source project dedicated to grid analysis, visualization and simulation, co-developed with RTE (Réseau de Transport d'Électricité), the French TSO.

The codebase provides a complete Knitro-based AC load flow solver that formulates the problem as a nonlinear optimization model with slack variables and a penalized objective. The solver is

¹<https://github.com/powsybl/powsybl-open-loadflow-knitro-solver>

²<https://www.powsybl.org/>

fully compatible with IIDM (iTesla Internal Data Model) networks and can be used as a drop-in replacement for Newton-Raphson in most simulation contexts.

To ensure reliability and stability, a suite of non-regression tests is in place:

- basic convergence tests on standard IEEE networks;
- comparison of load flow results between Newton-Raphson and Knitro on nominal cases;
- verification that the solver preserves the expected behaviour.

These tests run automatically and serve as a foundation for extending validation to a wider range of operating scenarios and network configurations.

Python interface. A custom Python interface to the solver is available in the `pypowsybl-open-loadflow-knitro-solver` repository³, which also provides a build example and a usage notebook. It exposes the resilient load flow on top of PyPowSyBl, which is convenient for prototyping, for post-processing the slacks and for reproducing the experiments described below.

Benchmark repository. The internal benchmark used for the experimental protocol of Section 7 is hosted in a dedicated repository⁴. It can be run locally, or launched through the GitLab CI on an internal runner. The test datasets being confidential, they are stored on Artifactory and retrieved at run time rather than versioned with the code.

6 Validation methodology

6.1 Comparison with Newton-Raphson on healthy networks

To assess the quality of the solution, we systematically compare the output of the resilient solver against a traditional Newton-Raphson run on the same healthy network, i.e. when both solvers converge. The quantities of interest — active and reactive injections, voltage magnitudes, current intensities and phase angles — are expressed in per-unit on a 100 MVA base. Fixed absolute tolerances are defined per variable to decide whether the results are equivalent.

Variable name	Symbol	Unit	Per-unit formula	Absolute tolerance for solvers comparison
Active power	P	MW	$\frac{P}{S}$	10^{-3}
Reactive power	Q	$MVar$	$\frac{Q}{S}$	10^{-3}
Target voltage	V	kV	$\frac{V}{V_{nom}}$	10^{-3}
Current Intensity	I	A	$\frac{\sqrt{3}V_{nom}}{S \cdot 10^3} I$	10^{-3}
Phase angle	θ	Degree °	-	10^{-3}

For per-unit calculations we used $S = 100MW$ as a reference value for the nominal apparent power.

Figure 3: Electrical quantities and tolerances used for the comparison.

³<https://github.com/Artelys/pypowsybl-open-loadflow-knitro-solver>

⁴<https://gitlab.artelys.lan/mdeboute/powsybl-open-loadflow-knitro-solver-benchmark> — internal access only.

As established in Section 3.4, on these cases the two formulations share the same solution set, so any discrepancy beyond tolerance signals an implementation or a conditioning issue, not a modelling one.

6.2 Stress tests: targeted perturbations

To validate robustness, we designed stress tests on IEEE networks (IEEE 9 to 300) by introducing controlled perturbations that push the system outside the feasible region of the standard formulation. These modifications are not random: they reproduce well-known physical sensitivities and real-life data defects. In each case we check both that a solution is returned and that **the active slacks point at the injected defect**.

6.2.1 Voltage regulation conflict

Real transmission datasets may contain inconsistent voltage controls, especially with remote voltage regulation. Two electrically close buses may carry different voltage targets, generating huge reactive flows that cannot be routed from distant generators through the network.

To emulate this, we introduce a localized perturbation creating a strong voltage conflict. We select a voltage-regulating generator, then a neighboring bus connected through a line with no voltage regulation, then a third bus neighboring the second one. If this third bus already hosts a generator, we modify its voltage target so that it slightly differs from the first one; otherwise we add a generator on the neighboring bus, regulating voltage at a different target. We then activate a remote voltage control for this generator on the second bus.

The line connecting the first two buses is modified as follows:

- series resistance set to zero;
- both shunt admittances set to zero;
- series reactance set to a very small positive value ($X = 10^{-4}$)⁵.

This creates an infeasible or ill-conditioned situation for Newton-Raphson without globally corrupting the rest of the network. We expect the resilient model to resolve the infeasibility through the voltage magnitude slack of one of the controlled buses.

6.2.2 Active power transmission

Pan-national security studies require merging national networks. This merging process can disrupt the global active balance, especially when the exchanged powers anticipated at the borders by the different TSOs are inconsistent. A network with a large imbalance between active generation and demand may be unsolvable, because it ends up requiring flows on lines that are not dimensioned for them.

Emulation: select a load and set its active power to 10% of the total network demand. Newton-Raphson should not converge; Knitro is expected to converge using the active balance slack of the bus hosting the load, allowing the end user to identify the origin of the issue. Note that this case is precisely the one where ρ_{PQ} of 5 is low, since $|\Delta_P|$ is large: the model is expected to accept the active slack rather than to distort the voltage plan.

⁵Keeping a small but nonzero reactance avoids two issues: it prevents numerical singularities in the Jacobian, and it ensures that PowSybl does not automatically merge the two connected buses, which happens below 10^{-8} p.u. The line thus remains in the model while creating a near short-circuit condition.

6.2.3 Reactive power transmission

Errors on shunt sections are frequent on real data and can significantly impair Newton-Raphson convergence. An erroneous shunt section position may let a bus produce a large amount of reactive power without any capability to export it towards the part of the network where it is needed.

Emulation: identify a bus where both a shunt and a generator are connected, make the generator regulate voltage locally, then modify the shunt section susceptance so that the shunt injects 1 GVar (using the generator target voltage as reference). The problem becomes infeasible and Newton-Raphson should not converge. With Knitro, we expect a feasible solution obtained through the voltage magnitude slack of that bus, leading to the identification of the defect.

6.3 Knitro stopping criteria

Parameter	Description	Default value	Used value
KN_PARAM_GRADOPT	An integer specifying the calculation mode of the gradients: 1 for user provides gradients, 2 for forward finite differences and 3 for central finite differences.	1	default
gradientUserRoutine	An integer specifying how to handle the Jacobian matrix: 1 for the dense option and 2 for the sparse option.	x	2
KN_PARAM_MAXFEVALS	Specifies the maximum number of function evaluations before termination. Values less than zero imply no limit.	-1	-
KN_PARAM_FEASTOL	Specifies the final relative stopping tolerance for the feasibility error.	1e-6	-
KN_PARAM_FEASTOLABS	Specifies the final absolute stopping tolerance for the feasibility error.	1e-3	-
KN_PARAM_FTOL	The optimization process will terminate if the relative change in the objective function is less than <u>ftol</u> for <u>ftol_iters</u> consecutive feasible iterations.	1e-15	-
KN_PARAM_FTOL_ITERS		5	-
KN_PARAM_INFEASTOL	Specifies the (relative) tolerance used for declaring infeasibility of a model.	1e-8	-
KN_PARAM_INFEASTOL_ITERS	The optimization process will terminate if the relative change in the feasibility error is less than <u>infeastol</u> for <u>infeastol_iters</u> consecutive infeasible iterations.	50	-
KN_PARAM_MAXIT	Specifies the maximum number of iterations before termination. 0 lets Knitro automatically choose a value based on the problem type. Currently Knitro sets this value to 10000 for LPs/NLPs and 3000 for MIP problems.	0	200
KN_PARAM_OPTTOL	Specifies the final relative stopping tolerance for the KKT (optimality) error.	1e-6	-
KN_PARAM_OPTTOLABS	Specifies the final absolute stopping tolerance for the KKT (optimality) error.	1e-3	-
KN_PARAM_XTOL	The optimization process will terminate if the relative change in all components of the solution point estimate is less than <u>xtol</u> for <u>xtol_iters</u> . If <u>xtol_iters</u> set to 0, Knitro chooses this value based on the solver and context. Currently Knitro sets this value to 3 unless the MISQP algorithm is being used, in which case the value is set to 1 by default.	1e-12	-
KN_PARAM_XTOL_ITERS		0	-
KN_PARAM_ALG	Indicates which algorithm to use to solve the problem: 0 (auto) let Knitro automatically choose an algorithm, based on the problem characteristics; 1 (direct) use the Interior/Direct algorithm; 2 (cg) use the Interior/CG algorithm; 3 (active) use the Active Set algorithm; 4 (sqp) use the SQP algorithm; 5 (multi) run all algorithms, perhaps in parallel.	0	-

Figure 4: Knitro stopping criteria.

6.4 Results

Newton-Raphson typically fails to converge on these instances, because of the conflict between incompatible voltage setpoints and the strong electrical coupling between the generators.

The resilient formulation consistently converges thanks to its slack variables on voltage magnitude, which allow conflicting voltage targets to be relaxed selectively. This demonstrates the ability of an optimization-based formulation to handle infeasibilities locally and to restore feasibility through minimal, explainable relaxations.

7 Experimental protocol

The following protocol isolates the effect of the two modelling decisions — dropping the ℓ_2 term, then calibrating the weights.

- (i) **Benchmark:** the internal benchmark repository introduced in Section 5 serves as the common test bed, locally or through the CI.
- (ii) **Baseline:** historical weights, $h(\varepsilon) = \varepsilon^2 + 3|\varepsilon|$, $\omega_P = \omega_Q = \omega_V = 1$.
- (iii) **Pure ℓ_1 , unit weights:** $h(s^+, s^-) = s^+ + s^-$, all weights at 1. Trivial to implement, isolates the sparsity effect of removing the quadratic term.
- (iv) **Pure ℓ_1 , calibrated weights:** Table 3 with the indicative values of Table 4.
- (v) **Comparison metrics,** on every instance: number of non-zero slacks, concentration of the violation (share carried by the largest slack), objective value, physical consistency of the returned point, agreement between the localization of the slacks and the injected defect, iteration count and solve time.
- (vi) **Sensitivity:** vary η and the parameters of ρ_{PQ} to map the arbitration between the three constraint families.
- (vii) **If needed:** revisit the engineering parameters, or the form of h .